

AIM

To study and write the PL/SQL programs using cursors.

Cursor:

A cursor in SQL is a temporary work area created in system memory when a SQL statement is executed.

Context area:

- Oracle creates a memory area known as context area, for processing an SQL statement
- A cursor is a pointer to this context area. PL/SQL controls the context area through a cursor.
- A cursor holds the rows (one or more) returned by a SQL statement.
- The set of rows the cursor holds is referred to as the active set

Two types of cursors:

1. Implicit cursors
2. Explicit cursors

Syntax:

CURSOR cursor_name IS select_statement;

Working with an explicit cursor involves four steps:

1.Declaring the Cursor

Declaring the cursor defines the cursor with a name and the associated SELECT statement.

Example: CURSOR c_customers IS SELECT id, name, address FROM customers;

2.Opening the Cursor

Opening the cursor allocates memory for the cursor and makes it ready for fetching the rows returned by the SQL statement into it.

Example: OPEN c_customers;

3.Fetching the Cursor

Fetching the cursor involves accessing one row at a time. Example: FETCH c_customers INTO c_id, c_name, c_addr;

4.Closing the Cursor

Closing the cursor means releasing the allocated memory. Example: CLOSE c_customers;

CODE

Program Using Implicit Cursor to Increase basic Of Each Customer By 500:

ENAME	EID	DOB	BASIC	DA	TOTAL
Arjun	101	15-MAY-00	25500.5	120	25120
Bala	102	22-NOV-99	30500	150	30150
Yokarajan	103	30-DEC-05	28500.75	130	28130

Ram	104	05-JUL-98	32500.25	140	32140
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```
SQL> declare
  2  total_rows number(2);
  3  begin
  4  update employee set basic = basic + 500;
  5
  6  if sql%notfound then
  7  dbms_output.put_line('no employees selected');
  8  elsif sql%found then
  9  total_rows := sql%rowcount;
 10  dbms_output.put_line(total_rows || ' employees updated');
 11  end if;
 12  end;
 13  /
```

4 employees updated

PL/SQL procedure successfully completed.

```
SQL> select * from employee;
```

ENAME	EID	DOB	BASIC	DA	TOTAL
-----	-----	-----	-----	-----	-----
Arjun	101	15-MAY-00	26000.5	120	25120
Bala	102	22-NOV-99	31000	150	30150
Yokarajan	103	30-DEC-05	29000.75	130	28130
Ram	104	05-JUL-98	33000.25	140	32140

Program Using Explicit Cursor to Fetch the Employee Details from the Table:

```
SQL> declare
  2  cursor c_employee is
  3  select ename, eid, dob, basic, da, total from employee;
  4  v_ename employee.ename%type;
  5  v_eid employee.eid%type;
  6  v_dob employee.dob%type;
  7  v_basic employee.basic%type;
  8  v_da employee.da%type;
  9  v_total employee.total%type;
 10  begin
 11  open c_employee;
 12  loop
 13  fetch c_employee into v_ename, v_eid, v_dob, v_basic, v_da,
v_total;
```

```

14      exit when c_employee%notfound;
15      dbms_output.put_line('ename: ' || v_ename || ', eid: ' ||
v_eid || ', dob: ' || v_dob || ', basic: ' || v_basic || ', da: ' ||
v_da || ', total: ' || v_total);
16  end loop;
17  close c_employee;
18  end;
19  /
ename: Arjun, eid: 101, dob: 15-MAY-00, basic: 26000.5, da: 120,
total: 25120
ename: Bala, eid: 102, dob: 22-NOV-99, basic: 31000, da: 150, total:
30150
ename: Yokarajan, eid: 103, dob: 30-DEC-05, basic: 29000.75, da:
130, total:
28130
ename: Ram, eid: 104, dob: 05-JUL-98, basic: 33000.25, da: 140,
total: 32140

```

PL/SQL procedure successfully completed.

Program using explicit cursor to calculate da & total:

```

SQL> declare
1  cursor c_employee is select EID, ENAME, BASIC from employee;
2  v_eid employee.EID%type;
3  v_ename employee.ENAME%type;
4  v_basic employee.BASIC%type;
5  v_da employee.DA%type;
6  v_netpay employee.TOTAL%type;
7  begin
8  open c_employee;
9  loop
10     fetch c_employee into v_eid, v_ename, v_basic;
11     exit when c_employee%notfound;
12     if v_basic < 15000 then
13         v_da := (v_basic * 25) / 100;
14     elsif v_basic <= 30000 then
15         v_da := (v_basic * 35) / 100;
16     else
17         v_da := (v_basic * 45) / 100;
18     end if;
19     v_netpay := v_basic + v_da;
20     update employee set DA = v_da, TOTAL = v_netpay where EID =
v_eid;
21     dbms_output.put_line('Updated: ' || v_ename || ' | DA: ' ||
v_da || ' | Net Pay: ' || v_netpay);
22 end loop;
23 close c_employee;

```

```

24  dbms_output.put_line('All rows updated successfully.');
```

```

25  exception
26  when others then
27      dbms_output.put_line('Error: ' || sqlerrm);
28  end;
30  /
```

```

updated: arjun | da: 9100.18 | net pay: 35100.68
updated: bala | da: 13950 | net pay: 44950
updated: yokarajan | da: 10150.26 | net pay: 39151.01
updated: ram | da: 14850.11 | net pay: 47850.36
all rows updated successfully.
```

PL/SQL procedure successfully completed

Create an explicit cursor to calculate results of students:

```
SQL> select * from student;
```

ROLLNO	STUD_NAME	DEPT	YEAR	MARKS
180	Yokarajan	IT	2	70
137	Ros	AIDS	1	65
131	rithi	MTS	2	50
134	gojo	FT	3	83
170	Viji	CSE	2	97

```

SQL> declare
2  cursor c_students is
3      select rollno, marks from student;
4
5  v_rollno student.rollno%type;
6  v_marks student.marks%type;
7  v_grade varchar2(1);
8  begin
9      open c_students;
10     loop
11         fetch c_students into v_rollno, v_marks;
12         exit when c_students%notfound;
13
14         if v_marks >= 90 then
15             v_grade := 'a';
16         elsif v_marks >= 80 then
17             v_grade := 'b';
18         elsif v_marks >= 70 then
19             v_grade := 'c';
20         elsif v_marks >= 60 then
21             v_grade := 'd';
22         else
```

```

23         v_grade := 'f';
24     end if;
25
26     update student
27     set grade = v_grade
28     where rollno = v_rollno;
29
30     dbms_output.put_line('updated rollno: ' || v_rollno || ',
marks: ' || v_marks || ', grade: ' || v_grade);
31 end loop;
32
33 close c_students;
34
35 dbms_output.put_line('all rows processed successfully.');
```

```

36 exception
37     when others then
38         dbms_output.put_line('error: ' || sqlerrm);
39 end;
40 /
```

updated rollno: 180, marks: 70, grade: c
updated rollno: 137, marks: 65, grade: d
updated rollno: 131, marks: 50, grade: f
updated rollno: 134, marks: 83, grade: b
updated rollno: 170, marks: 97, grade: a

all rows processed successfully.

PL/SQL procedure successfully completed.

MARK SPLITUP:

CONTENTS	MARKS ALLOTED	MARKS OBTAINED
Aim, Algorithm, SQL, PL/SQL	30	
Execution and Result	20	
Viva	10	
Total	60	

Result:

Thus, to study and write the PL/SQL programs using cursors is executed successfully.